

Decision Tree: Predicting Play from Outlook & Wind

Full Dataset

Student	Outlook	Wind	Play
1	Sunny	Weak	Yes
2	Sunny	Strong	No
3	Sunny	Weak	Yes
4	Sunny	Strong	No
5	Rainy	Weak	No
6	Rainy	Strong	No

Step 1: Entropy of the Target (Play)

Play = Yes: 2 Play = No: 4 (Total = 6)

$$Entropy(S) = - \left(\frac{2}{6} \log_2 \frac{2}{6} + \frac{4}{6} \log_2 \frac{4}{6} \right) \approx 0.918$$

Step 2: Information Gain for Outlook

Outlook	Count	Yes	No
Sunny	4	2	2
Rainy	2	0	2

$$Entropy(\text{Sunny}) = - \left(\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4} \right) = 1.0$$

$$Entropy(\text{Rainy}) = - \left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2} \right) = 0$$

$$IG(\text{Outlook}) = 0.918 - \left(\frac{4}{6} \cdot 1.0 + \frac{2}{6} \cdot 0 \right) \approx 0.251$$

Step 3: Information Gain for Wind

Wind	Count	Yes	No
Weak	3	2	1
Strong	3	0	3

$$Entropy(\text{Weak}) = - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right) \approx 0.918$$

$$Entropy(\text{Strong}) = - \left(\frac{0}{3} \log_2 \frac{0}{3} + \frac{3}{3} \log_2 \frac{3}{3} \right) = 0$$

$$IG(\text{Wind}) = 0.918 - \left(\frac{3}{6} \cdot 0.918 + \frac{3}{6} \cdot 0 \right) \approx 0.459$$

Step 4: Compare and Select Root

$$IG(\text{Outlook}) \approx 0.251 \quad IG(\text{Wind}) \approx \mathbf{0.459}$$

Root Node = Wind (higher Information Gain)
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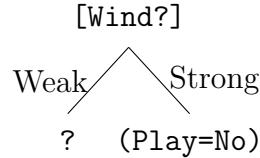
Step 5: Grow the Tree — Stage 1 (Root Only)

[Wind?]

Step 6: Split on Wind

Wind = Strong → all 3 samples → Play = No → **Leaf** (Entropy = 0)

Wind = Weak → 2 Yes, 1 No → still impure → needs further split



Step 7: Resolve Wind = Weak Branch

Remaining samples where Wind = Weak:

Student	Outlook	Play
1	Sunny	Yes
3	Sunny	Yes
5	Rainy	No

$$Entropy(\text{Wind} = \text{Weak}) = - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right) \approx 0.918$$

$$IG(\text{Outlook} \mid \text{Wind} = \text{Weak}) = 0.918 - \left(\frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 0 \right) \approx 0.918$$

Sunny → both Yes → **Leaf: Play = Yes** (Entropy = 0)

Rainy → only 1 sample, No → **Leaf: Play = No** (Entropy = 0)

Step 8: Final Tree — Stage 2 (Fully Grown)

